# Type MS · Explain chamber wall thickness and heart valves by their function

## Answer 1 of 6 · Question 7 part b · 5 marks

2023 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 7 | (a) | pulmonary artery; | | 1 |
| | (b) | • Y is thicker;<br>• blood pumped into aorta;<br>• takes/distributes blood to organs/tissues/whole body;<br>• requires high pressure<br>• thicker wall means more muscle for stronger contraction; | ORA<br>for each point | 5 |
| | (c) | any three of<br>• heart muscle deprived of oxygen;<br>• insufficient energy for muscle contraction/less aerobic respiration/(more) anaerobic respiration;<br>• heart stops pumping blood/pumps less blood;<br>• heart attack; | | 3 |
| | (d) | • transfers $CO_2$ (from muscle cells) to blood capillary;<br>• transfers oxygen from blood/capillary (to muscle cells);<br>• transfers glucose from blood/capillary (to muscle cells); | | 3 |

Total 12 marks

## Type MS · Explain chamber wall thickness and heart valves by their function

### Answer 1 of 6 · Question 7 part b · 5 marks

2023 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

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# Type MS · Explain chamber wall thickness and heart valves by their function

## Answer 2 of 6 · Question 1 part b.i · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.i in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) | B; (pulmonary artery)<br><br>*A it is not the aorta*<br>*C it is not the pulmonary vein*<br>*D it is not the vena cava* | | 1 |
| | | (ii) | A; (aorta)<br><br>*B it is not the pulmonary artery*<br>*C it is not the pulmonary vein*<br>*D it is not the vena cava* | | 1 |
| | | (iii) | in at top left and out through vessel;<br>in at top right and out through vessel; | | 2 |
| | | (iv) | valves;<br>prevent blood flowing other way/backflow; | | 2 |
| | (b) | (i) | chamber width in cm<br>P 1.0;<br>Q 0.2;<br>R 0.2;<br>S 1.5; | 1 correct = 1 mark<br>2 correct = 2 marks<br>3/4 correct = 3 marks | 3 |
| | | (ii) | An answer that makes reference to three of the following:<br>• The width corresponds to the amount of muscle;<br>• S pumps blood furthest distance/whole body;<br>• P pumps blood to lungs/short distance;<br>• Q/R only receives blood/doesn't have to pump it very far/to ventricles; | | 3 |

Total : 12 marks

## Type MS · Explain chamber wall thickness and heart valves by their function

### Answer 2 of 6 · Question 1 part b.i · 3 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

# Type MS · Explain chamber wall thickness and heart valves by their function

## Answer 3 of 6 · Question 1 part b.ii · 3 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part b.ii in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) | B; (pulmonary artery)<br><br>*A it is not the aorta*<br>*C it is not the pulmonary vein*<br>*D it is not the vena cava* | | 1 |
| | | (ii) | A; (aorta)<br><br>*B it is not the pulmonary artery*<br>*C it is not the pulmonary vein*<br>*D it is not the vena cava* | | 1 |
| | | (iii) | in at top left and out through vessel;<br>in at top right and out through vessel; | | 2 |
| | | (iv) | valves;<br>prevent blood flowing other way/backflow; | | 2 |
| | (b) | (i) | chamber width in cm<br>P 1.0;<br>Q 0.2;<br>R 0.2;<br>S 1.5; | 1 correct = 1 mark<br>2 correct = 2 marks<br>3/4 correct = 3 marks | 3 |
| | | (ii) | An answer that makes reference to three of the following:<br>• The width corresponds to the amount of muscle;<br>• S pumps blood furthest distance/whole body;<br>• P pumps blood to lungs/short distance;<br>• Q/R only receives blood/doesn't have to pump it very far/to ventricles; | | 3 |

Total : 12 marks

## Type MS · Explain chamber wall thickness and heart valves by their function

### Answer 3 of 6 · Question 1 part b.ii · 3 marks

2022 JAN · Paper 01 · extract 2 of 2 · find part b.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 4 of 6 · Question 5 part d · 2 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

| | | | |
|---|---|---|---|
| 5 (a) | • ACE inhibitor prevents/inhibits action of enzyme;<br>• prevents formation of angiotensin II;<br>• less( angiotensin ii) bind with receptors;<br>• Losartan blocks/binds to receptors;<br>• angiotensin ii cannot bind to receptor; | | 4 |
| (b) | • arteries/arterioles;<br>• muscles in walls;<br>• can contract/constrict;<br>• narrow lumen;<br>• smaller volume/space( for blood flow); | | 4 |
| (c) | • in normal/healthy kidney protein not found in urine;<br>• protein is a big molecule;<br>• should not pass through the glomerulus/basement membrane; | | 2 |
| (d) | • damage to glomerulus/basement membrane;<br>• stronger/intense/eq; <u>contractions</u>;<br>• (which causes) <u>muscle</u> to get thicker; | mp2 must be due to mp1 | 2 |
| (e) | • detoxification occurs in the liver;<br>• can't break down Losartan if impaired/accumulation /build-up in body/bloodstream/liver;<br>• results in (further) liver damage/liver failure/liver diseae; | | 2<br>**Total =14** |

## Type MS · Explain chamber wall thickness and heart valves by their function

### Answer 4 of 6 · Question 5 part d · 2 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | • A=rods/cones/retina; | | |

## Type MS · Explain chamber wall thickness and heart valves by their function

### Answer 5 of 6 · Question 1 part a.iv · 2 marks

2022 JAN · Paper 01 · extract 1 of 2 · find part a.iv in this mark scheme extract

| Question number | | | | | |
|---|---|---|---|---|---|
| 1 | (a) | (i) | B; (pulmonary artery)<br><br>*A it is not the aorta*<br>*C it is not the pulmonary vein*<br>*D it is not the vena cava* | | 1 |
| | | (ii) | A; (aorta)<br><br>*B it is not the pulmonary artery*<br>*C it is not the pulmonary vein*<br>*D it is not the vena cava* | | 1 |
| | | (iii) | in at top left and out through vessel;<br>in at top right and out through vessel; | | 2 |
| | | (iv) | valves;<br>prevent blood flowing other way/backflow; | | 2 |
| | (b) | (i) | chamber width in cm<br>P 1.0;<br>Q 0.2;<br>R 0.2;<br>S 1.5; | 1 correct = 1 mark<br>2 correct = 2 marks<br>3/4 correct = 3 marks | 3 |
| | | (ii) | An answer that makes reference to three of the following:<br>• The width corresponds to the amount of muscle;<br>• S pumps blood furthest distance/whole body;<br>• P pumps blood to lungs/short distance;<br>• Q/R only receives blood/doesn't have to pump it very far/to ventricles; | | 3 |

Total : 12 marks

## Type MS · Explain chamber wall thickness and heart valves by their function

### Answer 5 of 6 · Question 1 part a.iv · 2 marks

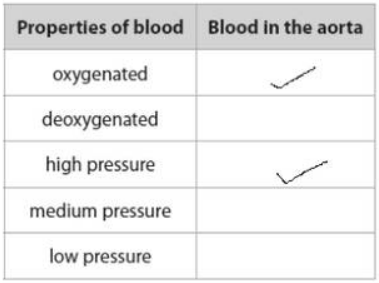
2022 JAN · Paper 01 · extract 2 of 2 · find part a.iv in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Explain chamber wall thickness and heart valves by their function

Answer 6 of 6 · Question 2 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

| Number | | | | | |
|---|---|---|---|---|---|
| 2 | (a) | (i) | left ventricle; | | 1 |
| | | (ii) | Two from:<br>• valve;<br>• prevents the backflow of blood;<br>• so the blood flows the correct way around the heart/in one direction; | Ignore names of valves | 2 |
| | | (iii) |  <br>Properties of blood / Blood in the aorta<br>oxygenated ✓<br>deoxygenated<br>high pressure ✓<br>medium pressure<br>low pressure | One mark removed for every additional incorrect tick. | 2 |
| | | (iv) | the lungs/pulmonary vein; | | 1 |
| | (b) | | Two from:<br>• it could be diseased/damaged/fatty deposits/blocked arteries;<br>• the tissue/blood type doesn't match;<br>• it has been removed from person for too long;<br>• wrong size; | accept named diseases<br>allow risk of rejection | 2 |
| | (c) | (i) | C platelets; | | 1 |
| | | (ii) | • fibrinogen is not converted to fibrin/fibrin not produced;<br>• so factor XIII has no substrate/fibrin cannot form the clot/mesh not formed; | | 2 |

Total 11 marks

## Type MS · Explain chamber wall thickness and heart valves by their function

### Answer 6 of 6 · Question 2 part a.ii · 2 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|